Radar-Led, Camera-Assisted BEV-Language Model for Driving Scene Understanding on nuScenes

Aamir Aziz Butt[1,2]
[1]PhD Scholar, Muslim Youth University, Islamabad, Pakistan
[2]Lecturer, Women University of AJ&K, Bagh, Pakistan
amir.butt12@gmail.com

Abstract—Radar-camera fusion is well established for 3D object detection, but its value for language-interfaced driving-scene understanding – a model that answers questions about a scene rather than only drawing boxes – remains largely untested, and existing driving question-answering benchmarks are built on camera and LiDAR alone. This work trains a small language model (Qwen2.5-0.5B, LoRA-adapted) to answer motion- and speed-related questions about nuScenes driving scenes from radar point clouds, camera features, and explicit per-object motion tokens, and evaluates radar's specific, measurable contribution using a majority-answer baseline, a wrong-frame control that distinguishes genuine sensor use from superficial pattern-matching, and a day/night/rain condition split. Under oracle (ground-truth) object boxes, radar's contribution is found to be specific rather than diffuse: every radar-containing model clears the majority baseline on pedestrian-motion counting, while camera alone does not, and every model's reliance on correct sensor content roughly doubles at night. Replacing oracle boxes with the project's own trained detector's real predictions, however, collapses the pedestrian-counting result while leaving car-motion and speed results intact. Seven independently implemented fixes targeting this gap – class-balanced object-token allocation, detection-threshold tuning, a confidence-score token feature, extended detector retraining with a full precision/recall sweep, attention-based pillar feature aggregation, auxiliary LiDAR-occupancy supervision, and cross-modal feature-and-response distillation from a LiDAR teacher detector nearly 3.3× better on pedestrians – all fail to recover it, evidence consistent with a sensor-level limitation of sparse automotive radar for low-radar-cross-section pedestrian targets rather than a fixable architectural defect.

I. Introduction

Automotive radar is established as a useful complement to camera for 3D object detection, where the two modalities' complementary strengths – camera's semantic richness and radar's direct velocity measurement and lighting-invariance – have been repeatedly shown to improve detection accuracy. Whether the same benefit transfers to a language-interfaced driving model – one that must answer natural-language questions about a scene, rather than only output bounding boxes – is comparatively untested: existing driving question-answering benchmarks are built exclusively on camera and LiDAR, with no radar baseline. This work asks a deliberately narrow question: can radar measurably improve a language model's ability to answer questions about a driving scene, and if so, specifically where, and does that improvement survive contact with the project's own, realistically imperfect object detector rather than only idealized ground truth? Section II positions this question against existing radar-camera detection work, camera/LiDAR-only driving question-answering benchmarks, and the handful of contemporary radar-language systems closest to this one. Sections III and IV then present the method and the resulting evidence, including an honest negative result for one of the two central findings once ground-truth boxes are replaced with real detections.

II. Related Work

A. Radar-Camera Fusion for Detection

Radar-camera fusion is well established for 3D object detection. CenterFusion [7] associates radar returns with camera-detected object centers via a frustum-based matching step, using the associated radar data to improve a camera-only detector's depth, rotation, and velocity estimates, reporting over a 12% improvement in nuScenes Detection Score over its camera-only baseline. RCBEVDet [8] is a more recent bird's-eye-view fusion approach for the same task. Both confirm that radar-camera fusion is worth doing for detection; this work's question is whether the same kind of benefit transfers to language, a model answering questions rather than drawing boxes. The pillar-based radar encoding used here follows the general pillarization approach introduced for LiDAR by PointPillars [9], and the CenterNet-style heatmap-plus-regression detection head follows the center-based formulation popularized for 3D detection by CenterPoint [10].

B. Driving Question-Answering Benchmarks

NuScenes-QA [5], the benchmark this work's evaluation protocol is built on, is constructed and baselined entirely on camera and LiDAR detections; radar is not part of its baselines. DriveLM [11] is a broader, graph-structured driving question-answering benchmark spanning perception, prediction, and planning questions, again built on camera and LiDAR with no radar. Neither benchmark tests whether a camera/LiDAR-centric approach degrades in the conditions – night, rain – where radar's physical advantages (direct velocity measurement, illumination

independence) should matter most; Section IV-C directly addresses this gap.

C. Radar-Only and Radar-Language Models

Two recent systems attempt something closer to a radar-language model. Radar4D-VLM [12] reasons over ten consecutive 4D-radar sweeps with no camera input at all, projecting object, scene, and kinematic tokens into several frozen language-model backbones; it reports that aligned language supervision gave no stable gain over its direct prediction heads. RLM [13] is trained contrastively on over 800,000 radar-caption pairs generated entirely in the CARLA simulator, rather than from real recorded sensor data, and produces free-form scene similarity rather than the exact-match, ground-truth-checked accuracy used here. Neither system combines real automotive radar with camera in the radar-led, camera-assisted framing this work uses, and neither reports a wrong-frame-style control to verify genuine sensor use.

D. Radar-Camera Scene Description and Robust Fusion

Two contemporaneous systems are closer in spirit. CRaFT-SD [14] is a sequential two-stage vision-language model: a LoRA-fine-tuned VLM reads a rendered range-azimuth map (converted to a pseudo-RGB image) and writes a structured text description of target position and clutter, which then conditions a second, pretrained VLM that also sees the camera image to produce a free-text operational-design-domain scene description. It is evaluated with BLEU score and target range/azimuth regression error (which it reports as substantially worse than a dedicated detector, 1.7 m vs. 0.11 m mean range error), rather than exact-match question-answering accuracy, and it carries no velocity or motion information at all – its radar representation is a static power map, not a point cloud with Doppler. RFusion4D [6] addresses a different problem, robust 3D detection under adverse weather, with a signal-domain denoising module and a confidence-aware temporal fusion encoder on high-resolution 4D imaging radar; it reports pedestrian AP of only 7.08% under adverse weather even with far better radar hardware and two orders of magnitude more training data than used here, independent evidence for the sensor-level pedestrian-detection ceiling this work documents directly in Section IV-D. Table I summarizes how this work's specific combination of choices differs from each.

This work's specific, deliberate combination – real nuScenes data rather than simulation, cheap sparse automotive radar rather than dense 4D radar, radar and camera together rather than radar-only, structured question-answering rather than free-form captioning, and a wrong-frame control methodology designed so that a model's answer sounding plausible cannot be mistaken for the model actually using the sensor – is not duplicated by any of the above.

III. Methodology

Fig. 1 shows the overall pipeline described in this section.

A. Dataset

Experiments are conducted on the nuScenes dataset [1], using the full v1.0-trainval split: 850 driving scenes (700 for training, 150 for validation), comprising 34,149 annotated keyframes collected across Boston and Singapore under varied weather and lighting conditions, including 84 training scenes tagged night and 138 tagged rain. Each keyframe provides six camera images, five radar sensors (front, front-left, front-right, back-left, back-right), and 3D bounding-box annotations with per-object velocity. This work restricts attention to the car and pedestrian categories, and uses the forward-facing camera as the sole visual input. Fig. 2 shows a representative keyframe: the front camera view alongside the corresponding accumulated radar point cloud and ground-truth boxes in the ego-centered BEV frame used throughout this work.

B. Radar Processing and Detection

1) Point Cloud Construction: Each radar sweep returns a sparse set of reflection points, each with position (x, y, z), ego-motion-compensated velocity (v_x, v_y), raw radial Doppler velocity v_r, and radar cross-section (RCS). A single sweep yields too few points per object for reliable detection, so the five most recent sweeps (approx. 0.5 s) from all five radars are accumulated per keyframe, following standard nuScenes practice. Each point is transformed into the ego vehicle's coordinate frame at keyframe time using the recorded ego pose and tagged with dt, the time elapsed since its own sweep, giving an 8-dimensional feature per point: $(x, y, z, v_x, v_y, v_r, \mathrm{RCS}, dt)$.

2) Pillarization: The accumulated point cloud is rasterized onto a 128×128 bird's-eye-view grid, 0.8 m per cell, covering a $102.4\,\mathrm{m} \times 102.4\,\mathrm{m}$ window centered on the ego vehicle. Points falling in the same cell ("pillar") are summarized by a learned PointNet-style pillar encoder (∼5,300 parameters), producing a 64-dimensional feature per occupied pillar and a $128 \times 128 \times 64$ bird's-eye-view tensor per keyframe.

3) Detector Architecture and Training: Pillar features pass through a U-Net backbone (∼1.85M parameters, downsampling then upsampling the spatial grid with skip connections), followed by CenterNet-style detection heads: a per-class Gaussian heatmap for object centers, plus regression heads for sub-cell offset, log-size, and sine/cosine heading. Training minimizes a focal loss on the heatmap and an L1 loss on the regression targets, using AdamW with a one-cycle learning-rate schedule and random left-right mirroring of each training frame. The detector is trained for 8 epochs on the 700-scene training split; Section IV-D reports that extending this to 20 epochs produces no measurable improvement. At inference, predictions are decoded via 3×3 local-maximum

TABLE I
Positioning relative to related work

Work	Radar type	Data source	Camera	Task	Attribution control
CenterFusion [7]	sparse	real	yes	detection	–
Radar4D-VLM [12]	4D	real	no	question answering	–
RLM [13]	sparse	CARLA simulation	no	free-form captioning	–
CRaFT-SD [14]	range-azimuth map	real	yes	free-form captioning	–
RFusion4D [6]	4D	real	yes	detection	–
This work	sparse	real	yes	question answering	wrong-frame control

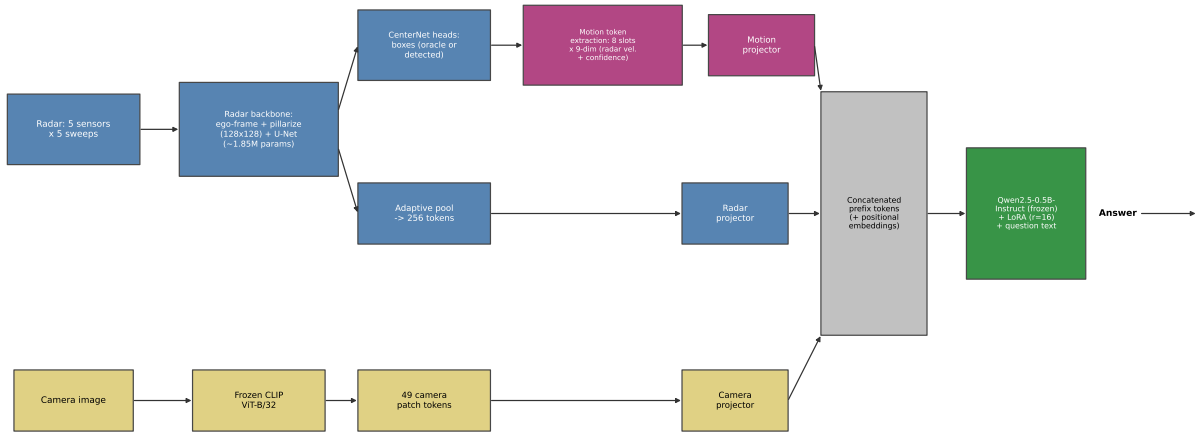


Fig. 1. Pipeline overview. Radar sweeps are accumulated in the ego frame, pillarized, and passed through a shared U-Net backbone that branches into a detection path (CenterNet heads → boxes → per-object motion tokens) and a dense-token path (adaptive pooling to 256 tokens); camera images pass through a frozen CLIP encoder. Each modality's tokens pass through their own lightweight projector before being concatenated into a single prefix fed to the LoRA-adapted language model, which generates the answer.

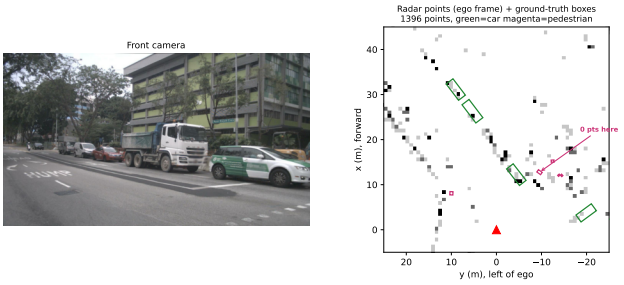


Fig. 2. A representative trainval keyframe. Left: front camera. Right: the corresponding 5-sweep-accumulated radar point cloud (greyscale, darker = more points per BEV cell) with ground-truth boxes overlaid. The highlighted pedestrian has zero radar points within its box radius, foreshadowing the detection-recall limitation examined in Section IV-D.

suppression on the heatmap, top-100 selection, and a configurable confidence threshold (default 0.3).

4) Detection Performance: On the 150-scene validation split the detector reaches AP@2m of 0.385 for cars and 0.120 for pedestrians (Table II) – a persistent asymmetry attributable to pedestrians' smaller radar cross-section and sparser returns, revisited directly in Section IV-D.

C. Camera Features

Camera input is encoded with a frozen, pretrained CLIP ViT-B/32 image encoder [2], used strictly as a fixed feature extractor with no fine-tuning. Each image is divided into a 7×7 grid of 49 patch tokens (768-dimensional each; the CLS token is discarded), cached per keyframe.

D. Explicit Object (Motion) Tokens

The dense 128×128 pillar grid is adaptively pooled to 16×16 (256 tokens) before being passed to the language model (Section III-E); at this resolution, small or sparse objects such as pedestrians can be attenuated or lost entirely. To preserve per-object information, an explicit object-token representation is constructed alongside the dense tokens: up to 8 slots per keyframe, split evenly between the nearest 4 cars and nearest 4 pedestrians to the ego vehicle, rather than the nearest 8 objects overall –

a shared pool would let the more numerous car detections crowd pedestrians out of the available slots. Each slot is a 9-dimensional feature vector:

$$[\,\text{is_car},\ \text{is_ped},\ x,\ y,\ v_x,\ v_y,\ \text{speed},\ \text{has_pts},\ \text{conf}\,]$$

where (x, y) is object position (normalized by 50 m); (v_x, v_y) is not a semantic label but the mean velocity of radar points falling within a radius of $\max(w, l)/2 + 1.0$ m of the object center (zero if no points fall within that radius); has_pts is a binary indicator of whether any such points exist; and conf is the detector's own classification score for that box, fixed at 1.0 when ground-truth ("oracle") boxes are used in place of detector predictions. Empty slots are zero-padded. Two box sources are used across experiments: oracle (ground-truth annotation) boxes, used to isolate the architectural question of whether object tokens help at all; and the project's own trained detector's predicted boxes, used in the realistic stress test of Section IV-D.

E. Language Model Architecture and Training

The backbone language model is Qwen2.5-0.5B-Instruct [3], adapted via Low-Rank Adaptation (LoRA, rank 16, applied to the query/key/value/output projections of every attention layer, $\alpha = 32$, dropout 0.05; 2,162,688 trainable parameters) [4], with pretrained weights otherwise frozen. Each sensor modality is converted into a sequence of soft prompt tokens via its own lightweight projector (LayerNorm → Linear → GELU → Linear, mapping into the LLM hidden dimension) plus a learned positional embedding indicating token rank or grid position. The resulting modality-specific token sequences – 256 radar tokens, 49 camera tokens, and/or 8 object tokens, depending on configuration – are concatenated and prepended as a prefix before the question text. In the full-fusion configuration, total trainable parameters (LoRA plus all active projectors) are 2,162,688 + 1,910,416. Training uses AdamW with separate learning rates for the projector(s) (10^{-3}) and LoRA parameters (2×10^{-4}), bf16 mixed precision, batch size 8, for 2 epochs on a randomly selected subsample of 60,000 training questions, evaluated on a disjoint 6,000-question validation subsample.

F. Question Generation

Two question families are used. NuScenes-QA [5] supplies the original benchmark's five question categories (existence, counting, object identification, status, comparison), templated from ground-truth scene annotations. To directly probe motion/speed reasoning, three additional question types are auto-generated from ground-truth object velocities: "how many cars/pedestrians are moving" (moving defined as speed > 0.5 m/s) and "is the fastest moving car's speed slow or fast" (fast defined as > 8.0 m/s; "none" if no car is moving). Majority-answer-per-question-type baselines for these three types are 0.195, 0.340, and 0.345 respectively.

G. Evaluation Protocol

All reported results use exact-match accuracy per question type. Three controls are used throughout:

Majority-Answer Baseline: accuracy from always predicting the single most frequent correct answer for a question type, with no model and no sensor data. Every reported result is interpreted relative to this floor.

Wrong-Frame Control: after normal training and evaluation, each model is re-evaluated with sensor tokens replaced by those of a different, randomly selected keyframe, while question text is unchanged. The resulting accuracy drop ("wrong-frame gap") is the primary evidence that a model reads the specific sensor content it was given, rather than relying on statistical regularities in question wording.

Condition Split: every accuracy and wrong-frame-gap figure is additionally computed separately for keyframes tagged clear, night, or rain, to test whether any sensor's measured advantage concentrates in the conditions where it has a physical reason to matter.

IV. Results

A. Detector Performance

TABLE II
Detector AP@2m, full trainval validation split (150 scenes)

Class	AP@2m
Car	0.385
Pedestrian	0.120

B. Modality Ablation on Motion/Speed Questions

All models in Table III and Fig. 3 are trained on the identical randomly sampled 60,000 training questions and evaluated on the identical 6,000-question validation subsample, using oracle object boxes, isolating the architectural question before addressing detector realism in Section IV-D.

TABLE III
Motion/speed question accuracy by modality (oracle boxes)

Model	Overall Acc.	Gap	Peds Moving
Majority baseline	–	–	0.340
Blind (no sensors)	0.513	–	0.34
Camera only	0.544	+0.053	0.33
Radar (dense) only	0.545	+0.041	0.36
Object tokens only	0.520	+0.055	0.43
Radar + object tokens	0.558	+0.075	0.45
Full fusion (+camera)	0.552	+0.070	0.47

The blind model's near-exact match to the majority baseline on all three question types (Fig. 4) confirms these questions cannot be answered from question-text statistics alone. Camera-only is the only model that fails to clear the majority baseline on pedestrian-motion counting (0.33 vs. 0.34), while every model containing radar information,

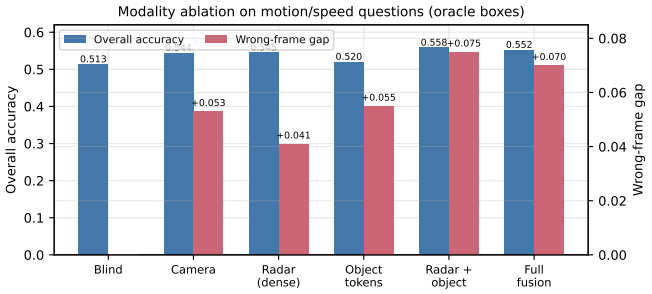


Fig. 3. Overall accuracy and wrong-frame gap across the six modality configurations, oracle boxes.

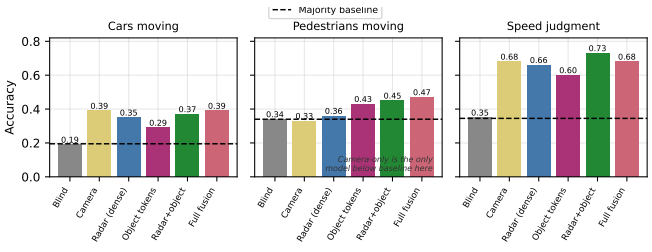


Fig. 4. Per-question-type accuracy. Camera alone is the only model that fails to clear the majority baseline on pedestrian-motion counting; every radar-containing model clears it.

in either dense or object-token form, clears it clearly (0.36–0.47). Radar's measured contribution is therefore not diffuse but specific: it concentrates on pedestrian-motion counting, consistent with the physical rationale that Doppler velocity is detectable regardless of an object's visual size. Dense radar features and explicit object tokens are complementary rather than redundant: the dense map's weakest question type is pedestrian counting, while the object tokens' weakest question types are car counting and speed; combined, radar + object tokens is the best or joint-best model on every metric, and adding camera on top yields no further overall improvement.

C. Day/Night/Rain Condition Split

Fig. 5 reports the wrong-frame gap from Table III's models, computed separately for the 4,660 clear-condition, 353 night-condition, and 987 rain-condition questions in the validation subsample.

Every model's wrong-frame gap is larger at night than in clear conditions, roughly doubling in most cases, indicating that every model relies more heavily on correct sensor content specifically at night. The two radar-containing object-token models show the largest night-condition gaps, exceeding camera-only's. Given that night accounts for only 353 of 6,000 validation questions, the exact magnitude of each figure carries an estimated uncertainty of roughly ±0.03; the consistency of direction across five independently trained models is the more statistically meaningful finding.

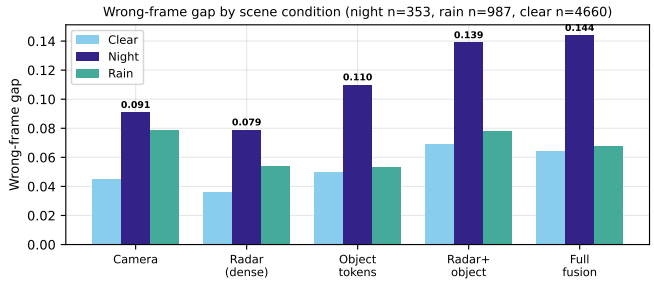


Fig. 5. Wrong-frame gap by scene condition. Every model's gap roughly doubles at night; the two radar+object models show the largest night-condition gaps of all.

D. Real-Detector Stress Test

Results above use oracle object boxes to isolate the architectural question. Table IV and Fig. 6 repeat the radar + object tokens evaluation with object tokens instead sourced from the project's own trained detector's predictions, testing whether the pedestrian-counting finding survives realistic, imperfect detection.

TABLE IV
Pedestrian-counting accuracy under four independently tested configurations

Configuration	Overall	Gap	Peds Moving
Oracle boxes	0.558	+0.075	0.45
Detected, balanced, 0.3 thr	0.519	+0.039	0.37
Detected, balanced, 0.15 thr	0.546	+0.049	0.36
Detected, +confidence, 0.3 thr	0.541	+0.048	0.36
Detected, +confidence, 0.15 thr	0.553	+0.050	0.35

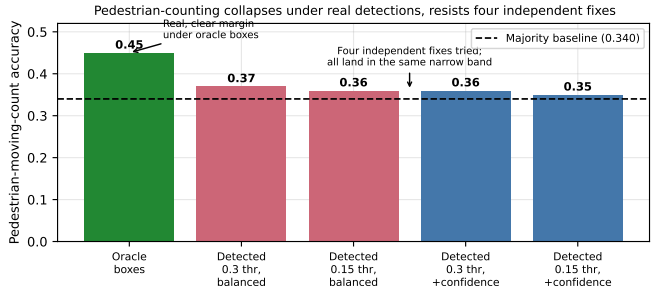


Fig. 6. Pedestrian-counting accuracy collapses when switching from oracle to detected boxes, and stays collapsed across four independently tried fixes.

Car-motion counting and speed judgment are robust to the switch from oracle to detected boxes (within approximately 0.02 of their oracle values across all four detected configurations). Pedestrian-motion counting is not: it falls from 0.45 to 0.35–0.37 across every detected configuration tested, a drop that persists despite three independently implemented interventions: (i) class-balanced slot allocation, ruling out cars crowding pedestrians out of the 8 available slots; (ii) a lower detection confidence threshold

(0.15 vs. the default 0.3), ruling out pedestrian detection recall as the limiting factor; and (iii) an explicit confidence-score feature added to each object token, letting the model learn to discount low-confidence detections. None recovers pedestrian-counting accuracy.

A fourth intervention – retraining the detector for 20 epochs (2.5× the original 8) with per-epoch checkpointing, plus a full precision/recall sweep across seven confidence thresholds (0.05–0.40) on the retrained checkpoint (Fig. 7) – was also tested. Both classes' validation AP plateaued by epoch 7–9 and did not improve further (car AP 0.385 → 0.378; pedestrian AP 0.120 → 0.109), and no threshold yields a usable pedestrian operating point (recall does not exceed approximately 0.27, reached only at a precision of approximately 0.04).

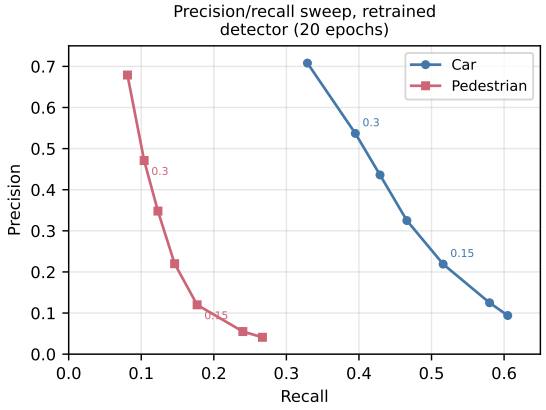


Fig. 7. Precision/recall sweep of the retrained detector. The car curve offers a usable operating range; no point on the pedestrian curve offers acceptable precision and recall simultaneously.

With four token-level and training-budget interventions exhausted, three further architectural interventions were tested, each going further than the last toward genuinely richer supervision. Fifth, the pillar encoder's max-pooling aggregation – which keeps only the single loudest point per channel per pillar, discarding the rest – was replaced with learned attention pooling, letting the network combine several points gradedly rather than keep only one. Sixth, an auxiliary training-time-only loss was added: nuScenes' own keyframe LiDAR sweep was rasterized onto the same BEV grid as a dense occupancy target, and a small head was trained to predict it from radar features alone, injecting outside information during training while leaving the detector's architecture and inference-time inputs unchanged. Seventh, a full LiDAR-only teacher detector (identical CenterNet-style architecture, LiDAR points in place of radar points) was trained and confirmed to be substantially stronger than the radar detector, then distilled into the radar student via feature distillation (cosine distance between student and frozen-teacher BEV feature maps) and response distillation (BCE between the student's heatmap logits and the teacher's sigmoid heatmap as a soft target), following CRKD [15].

TABLE V
Detector AP under three further architectural interventions

Detector	Car AP@2m	Ped AP@2m
Original radar (baseline)	0.385	0.120
Attention-pooling (fix 5)	0.383	0.117
LiDAR-aux occupancy (fix 6)	0.384	0.118
LiDAR teacher (reference ceiling)	0.720	0.400
Distilled radar, CRKD-style (fix 7)	0.383	0.116

Fig. 8 shows why the teacher is a meaningful reference ceiling rather than a marginal one: in a representative keyframe, the nearest pedestrian has zero radar points within its box radius (even after 5-sweep accumulation) against 18 LiDAR points in the same radius from a single sweep. Despite this real density gap and a teacher nearly 3.3× better on pedestrian AP, all three interventions land in the same narrow band as the first four (Table V) – statistically indistinguishable from the original detector. The distillation losses were not inert: feature-distillation loss fell from 0.352 to 0.274 and response-distillation loss from 0.099 to 0.092 over training, indicating the student genuinely moved toward the teacher's representations. This did not translate into additional detections. Because detector AP did not change under any of the three interventions, and every configuration in Table IV shows detector AP change and downstream pedestrian-counting change moving together, the full downstream language-model re-evaluation was not repeated for these three detectors.

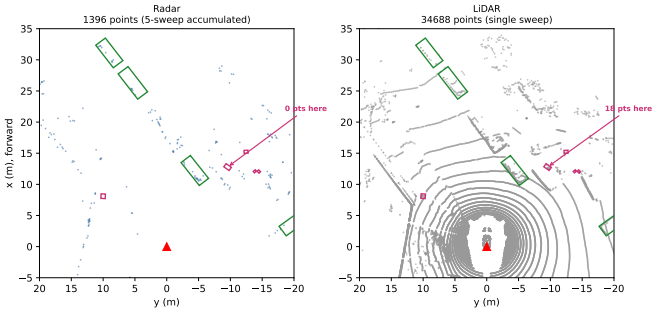


Fig. 8. Radar (5-sweep accumulated) vs. LiDAR (single sweep) points in the same keyframe. The highlighted pedestrian has zero radar points within its box radius against 18 LiDAR points in the same radius – a concrete case of the absent-information failure mode Table V points to.

All seven candidate fixes for the pedestrian-counting problem – spanning token design, detection thresholding, confidence-awareness, detector training budget, feature aggregation, auxiliary dense supervision, and real cross-modal distillation from a verified, substantially richer teacher – have now been tried independently, and all seven failed to recover it. The seventh is the strongest evidence of all: genuine knowledge transfer from a sensor that triples

pedestrian AP could not move the radar student's own detections. This is consistent with the limitation living in the radar student's input rather than in how that input is processed – if the sparse points needed to recognize a pedestrian were never captured, no amount of smarter aggregation, auxiliary supervision, or distillation pressure can manufacture detections from information that was never there.

E. Summary of Findings

Seven independently implemented and isolated interventions – object-token slot allocation, detection threshold, a confidence-score token feature, extended detector training with a full precision/recall sweep, attention-based pillar feature aggregation, auxiliary LiDAR-occupancy supervision, and cross-modal feature-and-response distillation from a verified, substantially richer LiDAR teacher – each targeting a specific hypothesis for the pedestrian-counting regression, all failed to close the gap between oracle and detected conditions. This convergent null result across seven architecturally unrelated interventions, including one that injects genuine outside information via a teacher nearly 3.3× better on pedestrian AP, is strong evidence against a fixable pipeline defect and consistent with a sensor-level limitation: pedestrians present a substantially lower radar cross-section and sparser, less consistent returns than vehicles, a limitation independently observed even with higher-resolution 4D imaging radar, two orders of magnitude more training data, and dedicated signal-domain denoising in contemporaneous work [6] (pedestrian AP of 7.08% under adverse weather, versus a 5.47% camera-only baseline). The results reported here therefore support a claim narrower than "radar improves driving-scene question answering broadly": radar's measured contribution is robust and specific to car-motion and speed reasoning under both idealized and realistic detection conditions, and additionally robust for pedestrian-motion reasoning only under idealized (oracle) conditions.

V. Discussion and Limitations

The night-condition result (Section IV-C) was obtained under oracle boxes only and has not yet been re-tested under the project's own detector's predictions; this is a direct item for future work, alongside multi-seed confidence intervals on the headline gaps. The pedestrian-counting ceiling documented in Section IV-D suggests a concrete design implication rather than a dead end: for vulnerable road users specifically, a camera-led, radar-assisted framing may be more appropriate than the radar-led framing supported here for vehicles.

VI. Conclusion

This work presents a controlled, ablation-based evaluation of radar's contribution to a language-interfaced driving-scene model, using a wrong-frame control and a day/night/rain condition split to separate genuine sensor use from superficial pattern-matching. Radar's contribution to car-motion counting and speed judgment is shown to be real and robust to realistic detector noise. Radar's contribution to pedestrian-motion counting is shown to be real under idealized conditions but is not yet demonstrated under realistic ones, after seven independently tried and isolated fixes – including cross-modal distillation from a LiDAR teacher nearly 3.3× better on pedestrian detection – a result best explained by a sensor-level limitation of sparse automotive radar for low-radar-cross-section targets, corroborated by an independent contemporary study.

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